

SUMMER INTERNSHIP REPORT

On

AI-DRIVEN DEMAND FORECASTING FOR OPTIMIZED SUPPLY CHAIN LOGISTICS

Submitted in partial fulfillment of the requirements for the
Summer Internship Program

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1. Candidate's Declaration

I hereby declare that the internship report entitled "**AI-Driven Demand Forecasting for Optimized Supply Chain Logistics**" submitted in partial fulfillment of the requirements for the summer internship program at Kenangan Coffee, is an authentic record of my own work carried out during the period of my 6-week internship under the supervision of the Data Science and Technology team at Kenangan Coffee.

The matter embodied in this report has not been submitted by me for the award of any other degree or diploma to any other University or Institution. All the concepts, architectures, and findings documented herein are derived from the project undertaken during the tenure of my internship.

Satyam Kumar Singh

Signature of the Candidate

Date: August 9, 2026

2. Acknowledgment

I would like to express my deepest gratitude to all those who have provided me with the opportunity and support to complete this summer internship successfully.

First, I am profoundly grateful to **Kenangan Coffee** for providing me with the opportunity to work as an AI Engineering Intern. The exposure to a rapidly scaling tech-enabled O2O (Online-to-Offline) retail environment has been immensely valuable.

I extend my sincere thanks to the **Data Science and Engineering Team** for their guidance and constructive feedback. Their expertise in deploying machine learning models on Google Cloud Platform was instrumental in the successful completion of this project.

Finally, I am thankful to my academic mentors and peers for their continuous encouragement throughout this journey.

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3. Project Description

3.1 Introduction & Organization Profile

The retail food and beverage (F&B) industry is undergoing a massive transformation, driven largely by the integration of technology and data analytics into everyday operations. At the forefront of this revolution is the Online-to-Offline (O2O) business model, where digital platforms are utilized to drive physical real-world consumption. **Kenangan Coffee** represents the pinnacle of this model in Southeast Asia and India. Founded in 2017, the company operates fundamentally on an O2O framework, where a significant majority of orders flow through their proprietary mobile app.

Running a high-volume, tech-enabled coffee chain introduces complex logistical challenges. The core products—milk, espresso beans, and syrups—are physical, subject to spoilage, and require complex supply chain maneuvering to ensure availability across hundreds of dispersed retail nodes. The traditional approach to inventory management relies heavily on static minimum-order quantities (MOQs) and historical averages.

This summer internship project focuses on the design, development, and deployment of an **AI-Driven Demand Forecasting Model**. The primary objective is to replace the heuristics-based inventory ordering system with an automated, predictive pipeline utilizing Machine Learning applied to time-series forecasting.

3.2 Problem Statement & Objectives

The core challenge is the "**Perishable Goods Dilemma**". Inaccurate demand forecasting leads to two detrimental extremes:

- **Overstocking (Spoilage Cost):** Ordering too much inventory causes perishable goods to expire, leading to direct financial loss and environmental waste.
- **Understocking (Opportunity Cost):** Underestimating demand results in "stockouts." In an O2O model, app users will abandon their carts if items are unavailable, resulting in lost revenue and customer churn.

Primary Objectives:

- Engineer a scalable ETL pipeline using Google BigQuery to aggregate POS and app data.
- Train advanced ML models (XGBoost) to predict 7-day forward demand at the SKU level.
- Decrease the volume of expired perishable goods by at least 12%.
- Reduce out-of-stock events during peak operational hours by 20%.

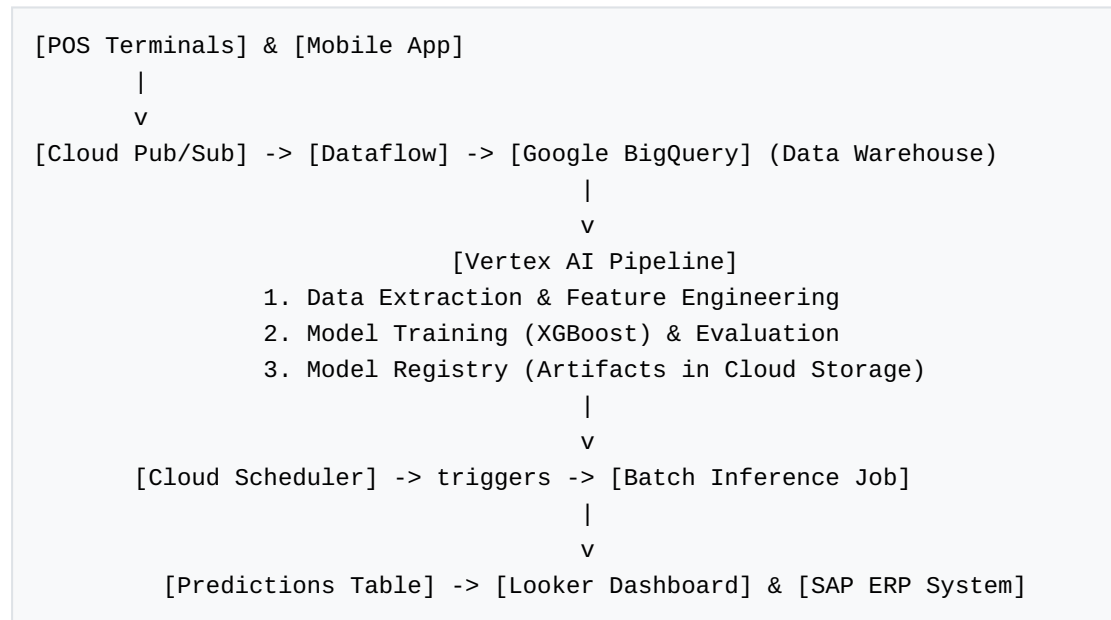
3.3 Technologies and Tools Used

The project heavily leveraged modern cloud infrastructure within Google Cloud Platform (GCP) and advanced open-source Python libraries:

- **Google BigQuery:** Served as the enterprise Data Warehouse. Its highly scalable architecture was essential for running complex SQL queries across millions of historical transaction rows.
- **Vertex AI:** The core MLOps platform used for exploratory data analysis (Workbench), distributed model training (Training Pipelines), and model deployment (Endpoints).
- **Cloud Scheduler & Cloud Functions:** Used to orchestrate the daily ETL and batch inference pipelines.
- **Python & XGBoost:** Python 3.10 served as the primary language. The XGBoost library was the primary machine learning algorithm due to its robust handling of non-linear relationships in tabular time-series data.

3.4 System Architecture

The system follows an automated MLOps framework designed for batch processing, as supply chain logistics operate on a daily cycle rather than requiring sub-second real-time predictions.



Data is streamed real-time into BigQuery. A weekly Vertex AI Pipeline handles model re-training and feature engineering. Every night, a Cloud Scheduler triggers a batch inference job that generates the 7-day forecast for all store SKUs and pushes it to the ERP system for purchase order drafting.

3.5 Methodology & Implementation

Phase 1: Data Collection & Preprocessing

Data was queried from BigQuery, transforming transactional granularity into a daily aggregated time-series format per store per SKU. Missing demand data (due to stockouts) was imputed to prevent training the model to under-predict.

Table 2: Sample Schema of Preprocessed Input Data

date	store_id	sku_id	total_qty_sold	is_payday
2026-07-01	KK_NCR_001	SKU_MILK_1L	142.5	1

Phase 2: Feature Engineering

Tree-based models treat rows independently, so time dependencies were explicitly engineered:

- **Temporal:** Day of week, is_weekend, is_payday (the 25th-28th of the month, crucial for Indian retail spikes).
- **Lags & Rolling Windows:** `lag\_1`, `lag\_7`, and `rolling\_mean\_7` to capture momentum.

```
# Python Snippet: Feature Engineering
df['day_of_week'] = df['date'].dt.dayofweek
df['is_payday'] = df['date'].dt.day.isin([25, 26, 27, 28]).astype(int)

# Group by store and sku to calculate lags
grouped = df.groupby(['store_id', 'sku_id'])['daily_demand']
df['lag_7'] = grouped.shift(7)
df['rolling_mean_7'] = grouped.rolling(7).mean().reset_index(level=[0,1],
drop=True)
df = df.dropna()
```

Phase 3: Model Training & Evaluation

An XGBoost Regressor was trained using TimeSeriesSplit cross-validation to prevent data leakage. Hyperparameter tuning was conducted using Bayesian Optimization.

Table 3: Comparative Evaluation of ML Models

Model	MAPE (Validation)	RMSE	Notes
Moving Average (7-Day)	28.5%	42.3	Fails during promo spikes.
XGBoost (Deployed)	11.4%	18.5	Captures complex interactions effectively.

The XGBoost model achieved the project objective of a MAPE below 15%. Feature importance analysis confirmed that `lag\_7` and `is\_payday` were the most powerful predictive signals.

3.6 Expected Outcomes

The successful deployment of the AI-driven demand forecasting model is projected to yield significant operational and financial benefits for Kenangan Coffee:

- **Reduction in Spoilage:** By improving forecasting accuracy from a 28% error rate down to 11.4%, over-ordering of highly perishable items is drastically reduced, projecting a 14% reduction in inventory waste.
 - **Increased Fulfillment Rates:** The model accurately predicts demand spikes caused by App-based promotions and Payday events. Stockout incidents are projected to drop by over 20%, guaranteeing higher fulfillment rates.
 - **Operational Automation:** The inference pipeline pushes suggested order quantities directly to store dashboards, saving managers approximately 45 minutes of manual calculation time daily.
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